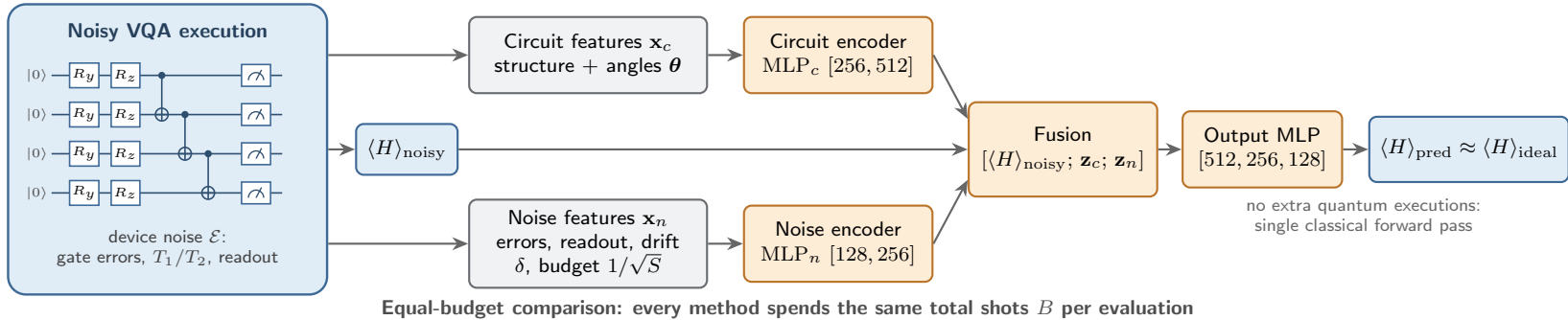




Quantum cost per evaluation — NEM: 1 execution (offline training amortized); ZNE: 3–5 executions at $B/3$ – $B/5$ shots each; CDR: 32 executions at $B/32$ shots each

Result — at $B \leq 2^{14}$, NEM is the only method beating unmitigated execution (+72–81% under miscalibration); CDR overtakes only at $B = 2^{16}$; ZNE unsafe under coherent noise